

Anthony Roca

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EDUCATION

University of North Carolina at Charlotte

Master of Science in Data Science & Business Analytics; GPA: 4.0

Charlotte, NC

Expected Dec 2026

Wake Forest University

Bachelor of Science in Computer Science; Minor in Mathematics

Winston-Salem, NC

July 2025

- Honors: Dean's List, Summer Management Program Completion with Distinction, School of Business Masters Scholarship

EXPERIENCE

Graduate Research Assistant

CHARM Lab, UNC Charlotte

November 2025 – Present

Charlotte, NC

- Conducting deep learning research for clinical-grade Alzheimer's detection using EEG/ERP signals, engineering feature extraction pipeline processing 100+ candidate features across time/frequency/spatial domains for peer-reviewed publication
- Engineered literature-driven feature identification system analyzing ERP features to address inter-subject variability and signal-to-noise ratio challenges in biomedical signal classification
- Developing hybrid CNN-LSTM architecture for 3-class classification problem (normal cognition, mild cognitive impairment, Alzheimer's disease) with goal of clinical deployment

Data Visualization Consultant

Todus Advisors (DSBA 5122 Industry Project)

Fall 2025

Charlotte, NC

- Developed interactive risk visualization platform for RiskTech startup enabling non-technical stakeholders to understand complex Bowtie diagrams through story-driven narratives, deployed to production at <https://dsba-bowtie.streamlit.app>
- Architected full-stack solution using React Flow and Streamlit processing 6+ risk barriers and scenarios, implementing expand/collapse, hover tooltips, and scenario toggle features reducing cognitive load in user testing
- Independently designed and implemented entire technical solution while coordinating team of 3 for presentation materials, receiving "EXCELLENT work" recognition from professor and RiskTech CEO with documentation approach adopted as future course requirement
- Built modular React components with state management for interactive diagram editing, integrated with Python backend for JSON persistence and narrative generation

Sports Data Scientist (Contract)

Carolina Core Football Club

July 2025 – Present

Charlotte, NC

- Built automated tactical scouting system processing 3,400+ events per match across 50+ games using StatsBomb API, delivering pre-match opponent analysis that directly informed coaching strategy and contributed to 8-1-3 playoff run versus 3-7-2 previous season
- Developed predictive models adapting advanced performance metrics (expected threat, value-added possession, passes per defensive action) to forecast opponent tactics, translating complex statistical insights into actionable recommendations driving 12+ tactical decisions
- Evaluated player performance versus contract value using data-driven analysis, uncovering roster inefficiencies that improved optimization by 56% while enhancing contract ROI, demonstrating measurable business impact
- Promoted from unpaid internship to paid contract position based on demonstrated value and tactical insights contribution to organizational turnaround

PROJECTS

Spotify Database ETL Pipeline | *Python, MySQL, SQL* | [GitHub](#)

Fall 2025

- Designed and implemented 3NF normalized MySQL database schema for 586K+ tracks and 1.1M+ artists from Spotify API, creating 6-table relational structure with proper foreign key constraints and indexing strategies
- Engineered modular ETL pipeline using Python orchestration with staging tables and incremental data transformation, processing large CSV files with command-line configuration for different environments

- Implemented complex analytical queries including outlier detection (1,934 tracks < 42s, 25,254 > 6.62min), genre analysis, and tempo distribution across decades with CTE-based query optimization
- Built comprehensive logging system with timestamps and error tracebacks, established Git version control workflow for reproducible database deployments

Steam Reviews - Big Data Mining | *Python, TF-IDF, Logistic Regression, ElasticNet, ALS*

April 2025

- Processed 7.8M+ reviews with TF-IDF and Logistic Regression, improving text classification accuracy from 42% to 65% and demonstrating robustness across temporal splits for sentiment analysis
- Built regression and recommendation models achieving MSE = 6.84 in collaborative filtering using Alternating Least Squares, while identifying data sparsity limitations that informed model selection decisions
- Modularized preprocessing and feature engineering pipeline in Python/scikit-learn for reproducibility and scalability, reducing end-to-end runtime by 35% across all modeling tasks through parallel processing

Urban Sound Classification - Deep Learning | *Python, CNN, Transformer, PyTorch*

May 2025

- Built hybrid CNN-Transformer architecture on UrbanSound8K dataset (8,732 audio clips), engineering MFCC and Mel-Spectrogram features with focal loss and augmentation to handle noise and class imbalance
- Achieved macro F1 = 0.584 under strict 10-fold cross-validation with reproducible results, debugging GPU bottlenecks and optimizing preprocessing pipeline for 40% faster inference
- Implemented attention mechanisms to capture temporal dependencies in audio sequences, outperforming baseline CNN-only approaches by 12% on minority classes

Penalty Kick Prediction - ML Research | *Python, Random Forest, LSTM, SHAP, OpenCV*

Oct – Dec 2024

- Scraped and labeled 160+ World Cup penalties, engineered pose landmarks from video using OpenCV, and automated ML pipeline with Random Forests and LSTMs, boosting prediction accuracy to 29-38% versus 16.7% baseline
- Recognized by faculty for innovation and recommended for startup competitions, highlighting applications in goalkeeper training optimization and performance analytics platforms
- Applied SHAP values for model interpretability to identify which body pose features most influenced penalty kick direction predictions

TECHNICAL SKILLS

Programming Languages: Python (expert), SQL (advanced), R, Java, JavaScript

Machine Learning & AI: Supervised Learning (Regression, Classification, Clustering), Ensemble Methods (Random Forest, XGBoost, Gradient Boosting), Deep Learning (CNNs, LSTMs, Transformers, U-Net), NLP (TF-IDF, Topic Modeling), Recommendation Systems

Data Engineering: Feature Engineering, ETL/ELT Pipelines, Data Preprocessing, Database Design (MySQL, SQLite), Dimensionality Reduction (PCA, UMAP), Imbalanced Data Handling (SMOTE, Resampling), PySpark

Tools & Frameworks: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, React, React Flow, Streamlit, Git, Docker, Tableau, Matplotlib, Seaborn

Statistical Analysis: Hypothesis Testing, A/B Testing, Cross-Validation, Experimental Design, Feature Importance (SHAP, Permutation)

Communication & Collaboration: Technical Documentation, Data Storytelling, Stakeholder Presentation, Cross-Functional Collaboration, Fluent Spanish